Structured notes: Section 1: growth and decay of current in an L-R circuit (Murugeshan 12.1, 12.2)

Murugeshan, *Electricity and Magnetism* (10th ed.), Ch. 12 §12.1 and §12.2 (Module 4, syllabus item 15–16). Figures are drawn for these notes from the book formulas; the book figure number is in each caption. Problems are not included.

12.1 Growth of current in an L-R circuit

Source: book-sources/m4-s12-1-growth.html, Murugeshan pp. 312–313 (printed). Syllabus item 15. Equations (1)–(4) are given in full.

Prerequisites

- Ohm's law and the emf of a source (Plus Two; pre-physics P1).
- First-order differential equations solved by separation of variables (Beyond; pre-math M4).
- Exponentials and natural logarithms (pre-math M3).

1. Setup

An inductance L and a resistance R are connected in series with a steady emf E through a key (Fig. 12.1). On closing the key the current rises from zero to a final value I_0. I denotes the current at time t.

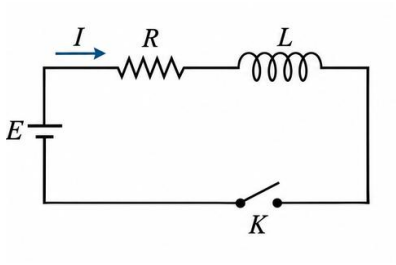


Fig. 12.1 (book): inductance L and resistance R in series with a cell E and key K. Schematic.

2. Loop equation

The induced back emf is $\varepsilon = -L\,dI/dt$. The loop equation is

$$E = RI + L\frac{dI}{dt} \qquad (12.1(1))$$

3. Final current

At the maximum, $dI/dt = 0$, so

$$E = RI_0 \qquad (12.1(2))$$

> **RECALL (PLUS TWO)** Ohm's law gives $I_0 = E/R$ for a steady current, which is the value the inductor allows once its back emf vanishes.

4. Variables separated

Substituting Eq. (2) into Eq. (1) gives $R(I_0 - I) = L\,dI/dt$, that is

$$\frac{dI}{I_0 - I} = \frac{R}{L}\,dt$$

5. Integration and boundary condition

Integration gives

$$-\log(I_0 - I) = \frac{R}{L}t + C \qquad (12.1(3))$$

The boundary condition $I = 0$ at $t = 0$ fixes $C = -\log I_0$, so $\log_e\,[(I_0 - I)/I_0] = -(R/L)t$.

6. Solution

$$I = I_0\left(1 - e^{-(R/L)t}\right) \qquad (12.1(4))$$

7. Time constant

The time constant is $\tau = L/R$. At $t = \tau$, $I = (1 - e^{-1})I_0 = 0.632\,I_0$. The current reaches $0.865\,I_0$ at 2τ and $0.950\,I_0$ at 3τ (Fig. 12.2).

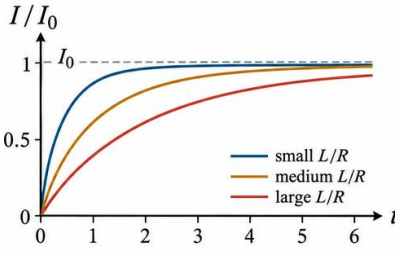


Fig. 12.2 (book): current I/I0 against time for low, medium and large L/R. Drawn from I = I0(1 − e^(−Rt/L)).

8. Worked example (printed p. 314)

For $R = 10\ \Omega$, $L = 0.5$ H and $E = 10$ V, $R/L = 20\ \text{s}^{-1}$. The time to reach 63.2% of I_0 is $t = \frac{\ln 2.717}{20} = 0.05$ s, which equals $\tau = L/R$.

12.2 Decay of current in an L-R circuit

Source: book-sources/m4-s12-2-decay.html, Murugeshan pp. 313–314 (printed), with Examples 1 and 2. Syllabus item 16. Equations (1) and (2) are given in full. Example 1 is worked in §12.1.

Prerequisites

- §12.1 of these notes: the loop equation and the time constant.
- Exponentials and natural logarithms (pre-math M3).

1. Setup

The cell is disconnected and the two ends of the circuit are joined. The emf is zero, and the inductor drives the current through R by its back emf.

2. Loop equation

$$0 = RI + L\frac{dI}{dt} \qquad (12.2(1))$$

Separating variables, $dI/I = -(R/L)\,dt$.

3. Solution

Integration gives $\log_e I = -(R/L)t + C$. The initial condition $I = I_0$ at $t = 0$ gives $C = \log_e I_0$, and so

$$I = I_0 e^{-(R/L)t} \qquad (12.2(2))$$

4. Interpretation

The decay time constant is L/R, the time in which the current falls to $1/e$ of I_0. The rate of decay is $dI/dt = -(R/L)I$. Growth and decay are complementary (Fig. 12.5).

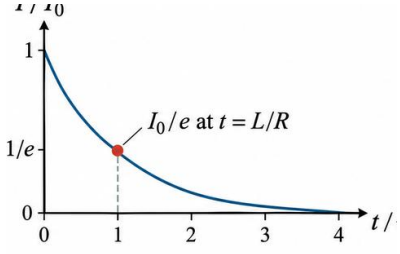


Fig. 12.3 (book): decay of current I = I0 e^(−Rt/L) against time. At t = L/R the current is I0/e.

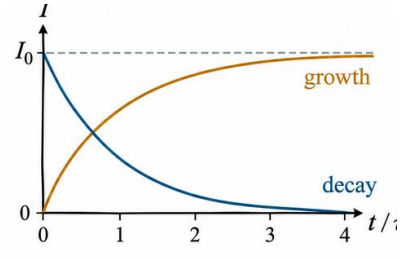


Fig. 12.5 (book): growth and decay curves shown together; they are complementary.

5. Worked example (printed p. 315)

$I_0 = E/R = 2$ A and $R/L = 10\ \text{s}^{-1}$, so $I = 2e^{-10t}$. At $t = 0.05$ s, $I = 1.213$ A. At $t = 0.2$ s, $I = 0.271$ A.